SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA02 **COURSE NAME: C Programming**

COURSE OUTCOME COVERED

CO3: *Solve problems for searching, sorting, and merging using array and string processing techniques. (BL3, BL4)*

Assessment Tool Weightage: 40%

QUESTIONS **Total Marks:100**

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I K.Kavya-192425213 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

90%

Maximum product subarray:

aim

to develop a program that finds the contiguous subarray with the maximum product while handling positive and nega-tive numbers efficiently.

problem understanding (20%)

the system must:

* identity the contiguous subarray with the highest product
* handle negative values carefully because multiplying two negative numbers gives a positive result
* track both maximum & minimum products dynamically

method selection (20%)

Dynamic programming approach is used

* maintain:

maxproud → maximum product ending at current index.

minproud → minimum product encoding at current index

* this method efficiently handles negative values using dynamic tracking

Interpretation of results (15%)

* the program successfully identifies the contiguous sub array with the maximum product
* Efficiently handles negative values, using dynam tracking.

87. Remove all occurance of character

Aim:

to create a program that removes all occurance of a specified character from a string efficiently.

problem understanding (20%)

the system must:

* Remove every occurance of a given character
* preserve the order of remaining character.
* Perform operation efficiently in-place.

method selection (20%)

* two-pointer technique is used
* one pointer reads character, another writes valid character

interpretation of results (15%)

* the program removes all specified character correctly
* original order of remaining character is preserved time complexity: O(n)

88. Union of two arrays

Aim:

to combine two arrays and produce a collection containing any unique elements

problem understanding

the system must:

* merge two datasets
* remove duplicate elements

preserve all unique data values

method selection

* Nested loop duplicate checking method is used.
* simple and suitable for moderate input series

interpretation of results

* the program successfully combines both arrays
* Duplicate values are removed
* ensures all unique elements are retained

89 count inversion in an array

Aim:

to determine how for an array is from being sorted by counting inversion pair

problem understanding

* the system must count pairs [i, j] where i<j and arr[i] >arr[j]
* efficiently compare ordering using optimized techniaue

method selection

* modified mergesent is used
* During merging, inversion are counted efficiently

interpretation of result

* the inversion count indicates has unsorted the array is higher inversion count means greater disorder time capacity : o(nlogn)

90 First Repeating element:

aim:

to identify the first repeating element in an array while preserving element order.

problem understanding:

the system must:

* Detect repeated elements

* Return the first repeating element based on occurance order

* efficiently track visited elements.

method selection:

* Nested loop comparison method is used

* checks occurance order carefully

interpretation of results:

* the program correctly identifies the earliest repeating element

* Reserve original order while checking repeatifing

* useful in log analysis and duplicate detection systems

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation